

Overvoltages and Capacitor Banks

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Abstract—Capacitor banks are widely used to correct a low power factor; however, this practice has drawbacks, since energizing a capacitor bank can produce overvoltages in the system. This work analyzes those overvoltages and the possible solutions. The simulation was carried out using Matlab and Simulink.

Index Terms—Capacitor banks, overvoltages, Simulink.

I. PROBLEM STATEMENT

Fig. 1 shows an electrical system supplying a factory that includes a motor control center and a low-voltage system feeding auxiliary services.

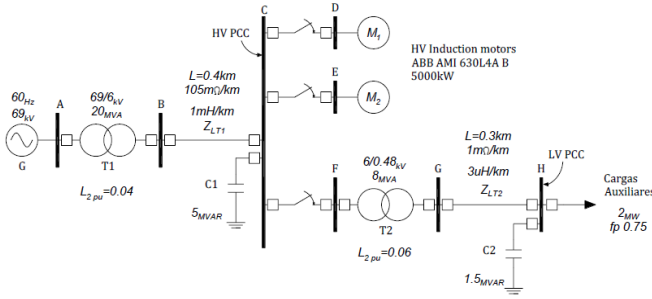


Fig. 1. Proposed electrical system.

A. Case 1

Once the HV motors are started and operating at full load, a capacitor bank used to compensate reactive power is connected a few seconds after the second motor starts. The low-voltage system is assumed to be disconnected. Determine the maximum transient overvoltage and overcurrent downstream of the capacitor-bank coupling point:

- 1) Show the maximum transient voltages and currents as a function of the point on the wave at which the capacitor bank is connected.
- 2) Compare the overvoltage and overcurrent waveforms of the best and worst possible cases.

B. Case 2

The low-voltage system is in operation and the capacitor bank is connected at the HV PCC. Using the simulation, determine the maximum transient overvoltage at the LV PCC and the transient overcurrent downstream of the LV PCC capacitor-bank coupling point:

- 1) Show the maximum transient voltages and currents at the LV PCC as a function of the point on the wave at which the capacitor bank is connected.
- 2) Compare the overvoltage and overcurrent waveforms of the best and worst possible cases.

C. Case 3

Consider now that the equipment connected as loads on the low-voltage side can withstand transient overvoltages of up to 1.8 times the nominal value. Propose two solutions to the problem, including figures with the transient currents and voltages seen at the end of the low-voltage line, before and after applying both methods.

II. BACKGROUND

The first step was to consider the magnitudes and data given in Fig. 1. Most of these were sufficient to configure the Simulink blocks, but the remaining elements—such as the motors—required investigating their model. With these parameters and applying (1), (2), and (3), the data needed to configure the simulation blocks were obtained.

$$\eta = \frac{P_{out}}{P_{in}} \quad (1)$$

$$|Z| = \frac{V^2}{S} \quad (2)$$

$$z = |Z| \cos(\theta) + j|Z| \sin(\theta) \quad (3)$$

For the capacitor banks, the configuration data are obtained from (4).

$$C = \frac{Q_C}{\omega V^2} \quad (4)$$

Since obtaining some results requires manipulating the Simulink data in Matlab, the “Structure with Time” format was selected under the “Logging” section of the “Configuration Properties” window of each Scope, so that the time and signal values are stored in a variable.

Finally, two fundamental times were considered in this work: a signal stabilization time and a coupling time. This means the elements are connected sequentially, as shown in Table I. In all three cases, the stabilization time lasts until the capacitor bank is connected; after that, the coupling time of bank 1 is varied to obtain the respective results.

TABLE I
CONNECTION OF THE ELEMENTS IN EACH CASE.

	Motor 1	Motor 2	LV system	Bank 1
Case 1	1st cycle	3rd cycle	—	5th cycle
Case 2	1st cycle	3rd cycle	5th cycle	7th cycle
Case 3	1st cycle	3rd cycle	5th cycle	7th cycle

III. ITERATION CODE

The results require the maximum overvoltages and overcurrents as a function of the capacitor-bank coupling point (in degrees), over one cycle in steps of 10° , together with a comparison between the best and worst possible cases.

This was implemented iteratively: each iteration produces a plot from which the maximum value is extracted, after which the coupling time of the capacitor bank is changed. A `for` loop was used with a variable t sweeping all the coupling points. Inside the loop, the simulation is executed with the `sim` function; beforehand, the `set_param` function assigns the new coupling point to the corresponding Simulink block from Matlab, specifying the file, the block, and the parameter to be varied, accounting for both the stabilization time and the coupling time. An array then stores all the maximum values, taken as the absolute value of the data. Finally, the loop is closed and, for plotting, the times are converted from seconds to degrees and the peak magnitudes to per-unit values.

IV. RESULTS

A. Case 1

1) *Maximum transient voltages and currents:* Connecting capacitor bank 1 changes the voltage and current signals, producing Fig. 2. The transients reach between 1.5 and 2.6 p.u. with respect to the nominal voltage, which at this point is 6 kV. The largest transient occurs at two coupling points, 80° and 260° after the stabilization time, while the smallest occurs at 0° , 180° , and 360° .

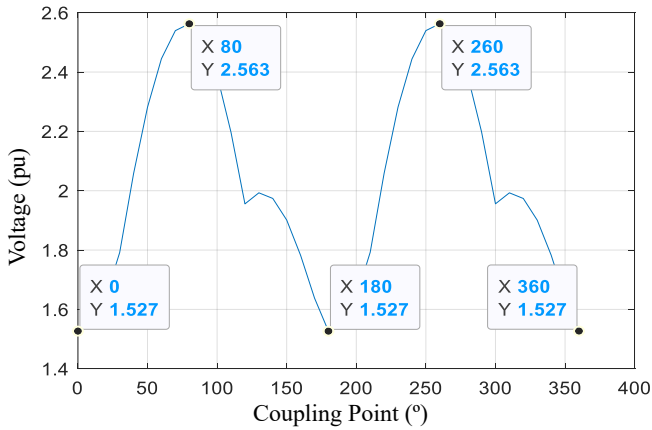


Fig. 2. Maximum transient voltages as a function of the coupling point, case 1.

The corresponding maximum transient currents are shown in Fig. 3. The currents range between 0.8 and 1.1 p.u. with

respect to the nominal current, which—from the secondary voltage and the transformer rating—is about 3333 A. The largest overcurrent occurs at the same coupling points as the overvoltage (80° and 260°), and the smallest at 0° , 180° , and 360° .

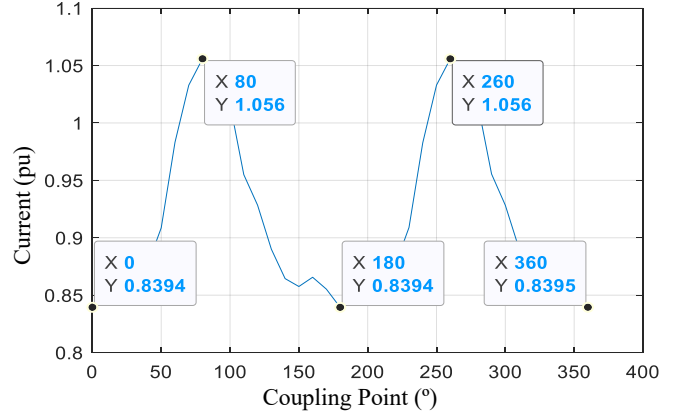


Fig. 3. Maximum transient currents as a function of the coupling point, case 1.

2) *Best- and worst-case signal comparison:* Fig. 4 compares the best- and worst-case maximum transient voltages, contrasting coupling at 0° (best case) with 80° (worst case). The transient in the best case is smoother, whereas in the worst case the signal is more distorted. The difference with respect to the 180° (best) and 260° (worst) points is that those peaks occur in the negative half-cycle of the wave.

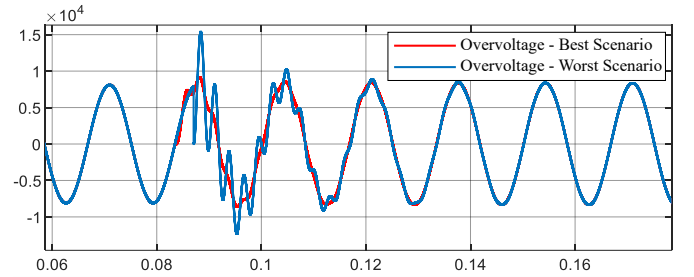


Fig. 4. Overvoltage, best vs. worst case. Case 1.

The same comparison for the currents, at 0° (best) and 80° (worst), is shown in Fig. 5. The current magnitudes are not as large and their transients are not as severe as those of the voltage. As with the overvoltages, the 180° and 260° points are the peaks occurring in the negative half-cycles, keeping the same magnitude but differing in where they occur.

B. Case 2

In this case two transients occur: one when the low-voltage system is connected and another when capacitor bank 1 is energized. Since only the latter is of interest, the iteration code for case 2 must take the absolute maxima starting from a given data point—here from sample 23150, i.e., from 0.11575 s onward.

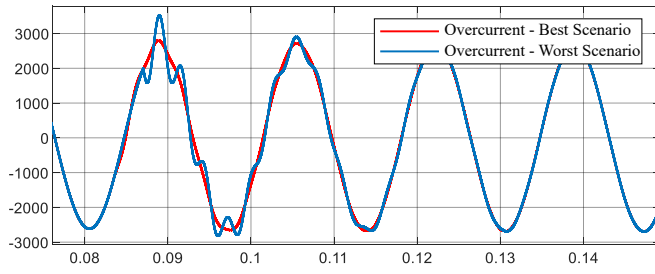


Fig. 5. Overcurrent, best vs. worst case. Case 1.

1) *Maximum transient voltages and currents:* Applying the same iteration code gives Fig. 6 for the transient overvoltages. The overvoltages reach between 1.48 and 3.5 p.u. with respect to the nominal value, now 480 V. As in case 1, the largest overvoltages occur at 80° and 260° and the smallest at 0° , 180° , and 360° after the stabilization time, with 0° being the best case overall.

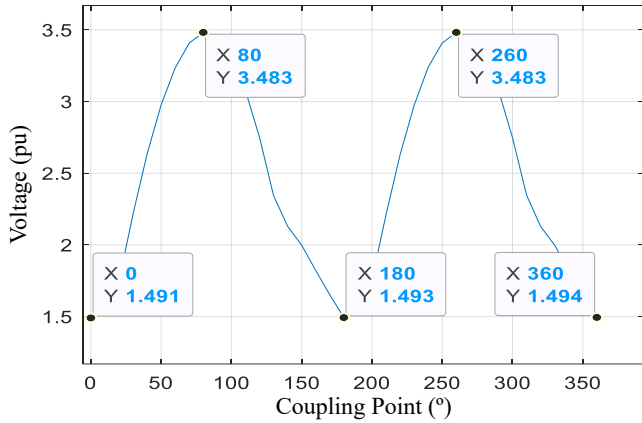


Fig. 6. Maximum transient voltages as a function of the coupling point, case 2.

The overcurrents are shown in Fig. 7, ranging between 0.46 and 0.6 p.u. with respect to the nominal value, which at this point is approximately 16667 A. Here the behavior differs slightly: the largest maximum current occurs at 90° in the positive half-cycle and 270° in the negative half-cycle, while the smallest occurs at 0° , 180° , and 360° .

2) *Best- and worst-case signal comparison:* Fig. 8 compares the best and worst cases for the voltages. Both the best- and worst-case overvoltages occur in the positive half-cycle, since in the negative half-cycle the smallest transient is slightly larger than its positive-half-cycle counterpart. The figure also shows the large difference between the two coupling points, the worst-case transient lasting longer than the other—which in practice can cause considerable damage.

The overcurrent comparison is shown in Fig. 9, simulating the 0° and 90° points for the best and worst cases, respectively. Although Fig. 7 indicates that the largest overcurrent occurs in the negative half-cycle, the positive-half-cycle data were used

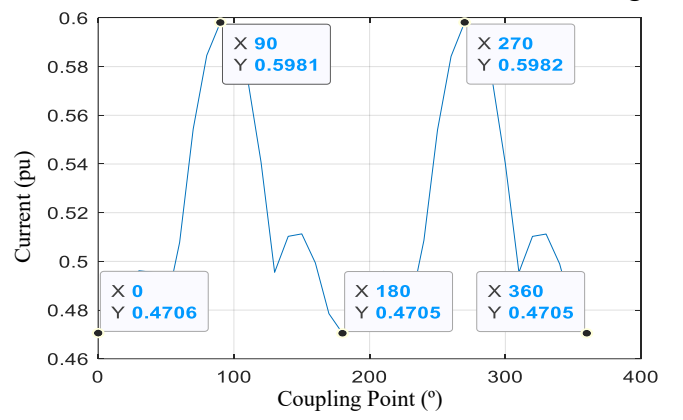


Fig. 7. Maximum transient currents as a function of the coupling point, case 2.

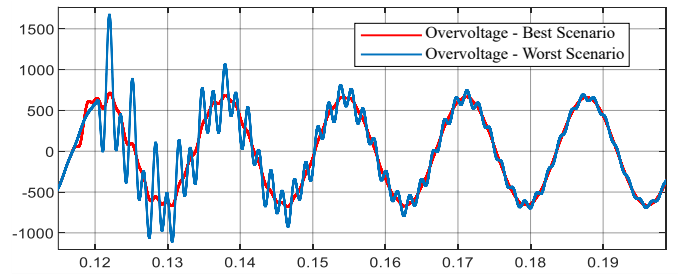


Fig. 8. Overvoltage, best vs. worst case. Case 2.

in order to contrast the overcurrent and overvoltage signals at the same points.

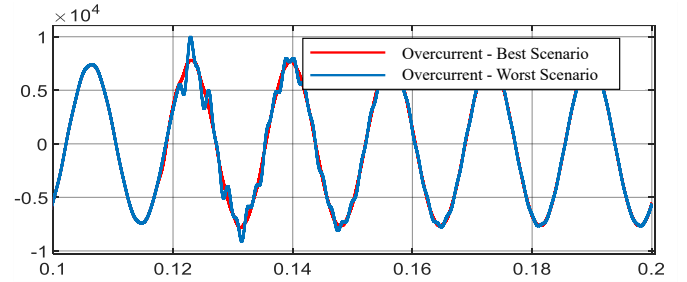


Fig. 9. Overcurrent, best vs. worst case. Case 2.

C. Case 3

Case 3 requires two solutions for the system, given that the auxiliary loads withstand only 1.8 times the nominal value. The solutions chosen were a high-energy surge arrester and a pre-insertion resistor.

1) *Solution 1: Surge arrester:* Surge arresters protect an installation against overvoltages. From Fig. 6, the worst-case overvoltage occurs when the bank is coupled at 80° and 260° after the stabilization time; the point in the positive half-cycle was used here.

The arrester was simulated with the “Surge Arrester” block, specifying the reference voltage and current and leaving the

remaining parameters at their default values. With the reference voltage set at 1.8 p.u. of nominal, the reference current was sized by trial and error, arriving at 25000 A. Applying these data to the worst-case scenario gives Fig. 10.

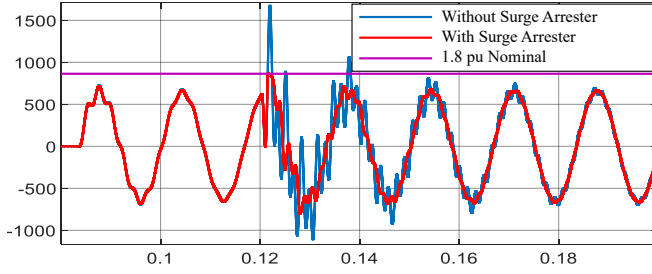


Fig. 10. Overvoltage in the worst case, with and without a surge arrester.

The overvoltage now reaches the desired limit without completely altering the waveform, since the signal restabilizes after some time. This result was obtained through trial-and-error sizing; the transient could be reduced further by increasing the reference current or the number of arrester columns.

The corresponding overcurrent is shown in Fig. 11. As has been the pattern throughout this work, the voltage variation acts directly on the current, so applying the arrester also reduces the current.

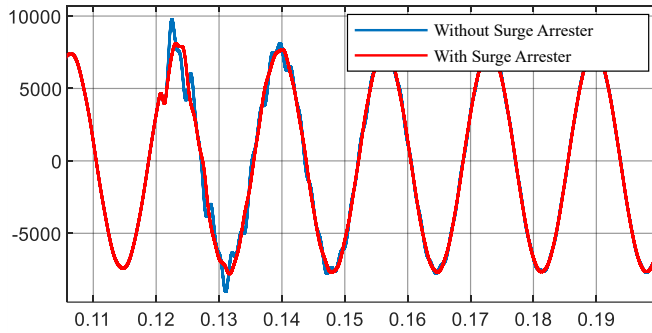


Fig. 11. Overcurrent in the worst case, with and without a surge arrester.

To verify the effectiveness of this solution, Fig. 12 shows the maximum per-unit voltages reached at every coupling point, all complying with the 1.8 p.u. limit.

2) *Solution 2: Pre-insertion resistor*: Pre-insertion resistors also reduce overvoltages. They consist of a resistor placed in series with the line through an initially open switch, which closes at the instant the transient occurs so that the current and voltage are reduced.

In this simulation, the switch closes at the instant of the worst possible overvoltage, inserting the resistor. Trial and error determined a value of approximately 364 m Ω for the overvoltage to fall just to the limit, as seen in Fig. 13.

The pre-insertion resistor produces a notable reduction in the transient voltage, and the signal stabilizes faster than with the arrester; however, the voltage settles slightly below its initial value. As with the previous solution, the resistance can

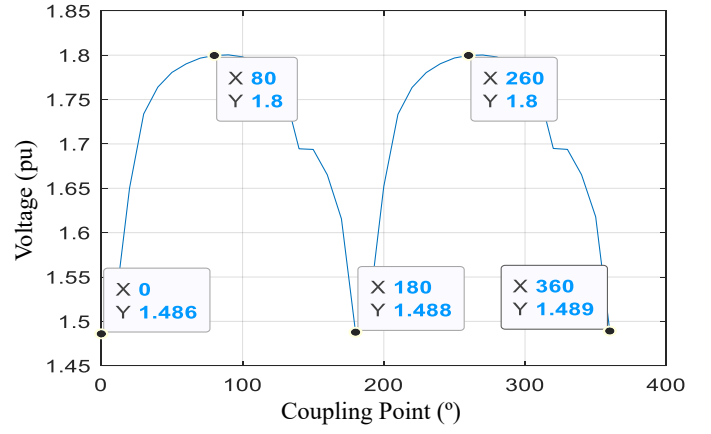


Fig. 12. Overvoltages with a surge arrester as a function of the capacitor-bank coupling point.

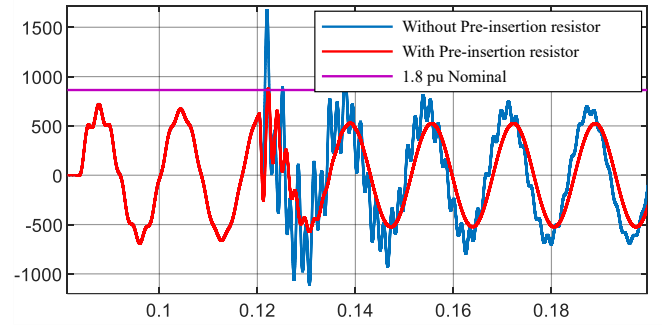


Fig. 13. Overvoltage in the worst case, with and without a pre-insertion resistor.

be varied to reduce the transient further, bearing in mind that the signal amplitude will also decrease.

The overcurrent for this solution is shown in Fig. 14. The current after the pre-insertion resistor also falls below its original value, because the resistor is not disconnected; even so, the difference in overcurrent amplitude at the worst-case point, before and after inserting the resistor, is notable.

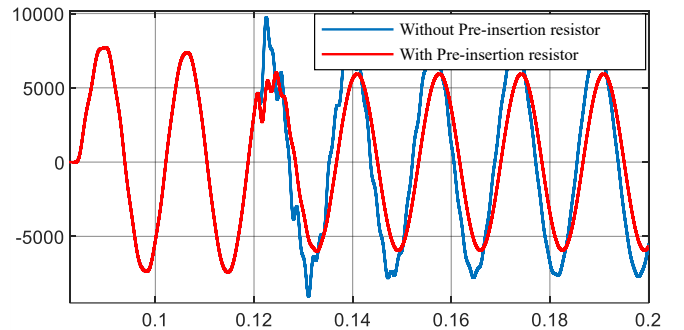


Fig. 14. Overcurrent in the worst case, with and without a pre-insertion resistor.

To verify the effectiveness of the pre-insertion resistor, Fig. 15 shows that the case-2 overvoltages are considerably

reduced; at the points where the value remains constant, the overvoltages are levelled out by the resistor.

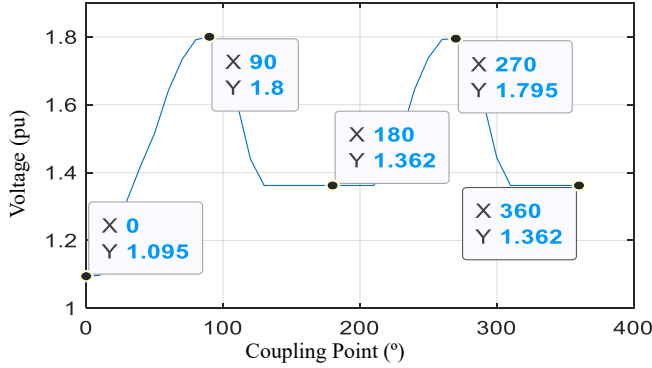


Fig. 15. Overvoltages with a pre-insertion resistor as a function of the capacitor-bank coupling point.

V. CONCLUSIONS

The analysis and simulation of the different cases showed how important it is to guard against overvoltages, which on their own can cause considerable damage. Their main causes include lightning, the energization of lines and feeders, and capacitor-bank switching. The latter is a double-edged sword: capacitor banks are used to improve the power factor, but they can also produce brief transient voltages and currents that are harmful to equipment. This work simulated what happens when a capacitor bank is connected to a distribution system supplying a factory.

A potential effect of adding a capacitor bank to improve the power factor is that the impact of the transients on the user's equipment can increase considerably. This is confirmed by comparing the per-unit values of cases 1 and 2: the per-unit voltage range in case 1 is much smaller than in case 2, as seen by contrasting Fig. 4 with Fig. 8. The more loads connected to the system, the greater the impact of the transient.

Capacitor banks are therefore both useful and risky. Rather than avoiding them, the best approach is to connect the bank at the instant where the transient is smallest—which cannot always be determined—or, more simply, to implement mitigation measures. This work used pre-insertion resistors and high-energy surge arresters, and both proved to be good options for reducing the maximum transients and protecting the equipment.

Comparing Fig. 10 and Fig. 13 reveals some differences. With the arrester, the voltage signal remains distorted for some time before restabilizing, whereas with the pre-insertion resistor the transient decreases as soon as the resistor is activated, holds briefly, and then the signal stabilizes. In the pre-insertion simulation the voltage decreases after the transient disappears; this is because the resistor should be connected for only a short interval and then switched out again, and the ideal switch used here could not reproduce that timed opening. The relevant outcome is that both solutions reduced the transients.

Finally, in this system the smallest maximum transient voltages consistently occurred when the capacitor-bank coupling signal passed through zero, while the largest occurred near each quarter-cycle, that is, close to 90° and 270° . A transient can also be more dangerous depending on where and when it occurs and on how many loads it can affect. Overvoltages are undesirable events in an electrical system, and techniques such as those analyzed here have been developed to attenuate them, maintaining good power quality and guaranteeing a stable, functional service.

REFERENCES

- [1] MathWorks. [Online]. Available: <https://matlab.mathworks.com/>